AP CALCULUS AB

Unit 1 • Lesson 1-5 • Guided notes

Lesson 1-5 • IVT and EVT

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

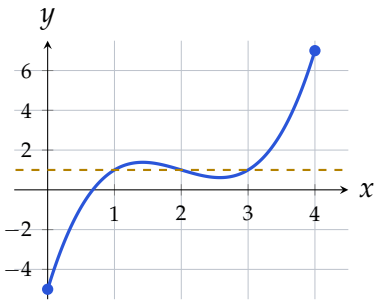
- State the Intermediate Value Theorem and the Extreme Value Theorem, with their hypotheses.
- Use the IVT to justify that an equation has a solution.
- Use the EVT to justify that absolute extrema exist, and find them from graphs and tables.

Vocabulary. Intermediate Value Theorem (IVT), Extreme Value Theorem (EVT), closed interval, absolute maximum, absolute minimum, hypothesis, conclusion.

1. Intermediate Value Theorem

IVT

If f is _____ on the _____ interval $[a, b]$ and k is any number _____, then there is at least one c in $[a, b]$ with _____.



Example 1

Show that $x^3 + 2x = 5$ has a solution in $[1, 2]$.

Example 2 • Table

W is continuous with $W(0) = 8$, $W(2) = 3$, $W(5) = 6$, $W(9) = 1$. What is the least number of solutions of $W(t) = 4$ on $[0, 9]$?

2. Extreme Value Theorem

EVT
If f is _____ on the closed interval $[a, b]$, then f has both an _____ and an _____ value on $[a, b]$. They occur at _____ or at points where the graph turns (critical points).

Example 3

Find the absolute maximum and minimum of $f(x) = x^2 - 4x + 5$ on $[0, 3]$. Why must they exist?

Example 4 • When the theorems do not apply

Explain why the conclusion can fail: (a) $g(x) = \frac{1}{x}$ on $[-1, 1]$, IVT with $k = 0$; (b) $h(x) = \frac{1}{x}$ on $(0, 1]$, EVT.

AP exam tip
An IVT or EVT justification names the hypothesis ("f is continuous on $[a, b]$ because…"), the values used, and the conclusion.

Check your understanding

1. f continuous, $f(1) = 4$, $f(3) = 4$. Does the IVT guarantee $f(c) = 0$ on $[1, 3]$? _____.
2. Does a continuous function on $(0, 5)$ always have an absolute maximum? _____.